

Notes: Harrison and Kreps (QJE, 1978)

Speculative Investor Behavior in a Stock Market with Heterogeneous Expectations

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1 Big picture

One-sentence summary. This note summarizes Harrison and Kreps (1978), which builds a dynamic stock-pricing model with heterogeneous beliefs, resale opportunities, and short-sale constraints, showing that the current price can exceed every investor's own present value of future dividends because ownership includes a valuable resale option.

Research question. What is the equilibrium price of a stock when investors disagree about future dividends, cannot sell the stock short, and expect that markets will reopen in the future?

Main answer. A stock price can contain a **speculative premium**. The premium arises because the current buyer values not only the dividends but also the right to resell the stock later to another investor class that may become more optimistic after future information arrives. Therefore, the price may exceed the dividend value assessed by any single investor class under a buy-and-hold rule.

Core mechanism.

- Investors have heterogeneous subjective probability assessments over future states.
- Short sales are ruled out, so pessimists cannot fully express negative beliefs by selling short.
- The stock is in positive net supply; in each state it is held by the class that values it most.
- A buyer may be willing to pay more than her own dividend valuation because she expects to resell to a different class in a future state.
- Dynamic resale value feeds back into today's price through a fixed-point condition.

Economic intuition. The stock is not merely a claim to a dividend stream. It is also a claim to the ability to trade the dividend stream in the future. When beliefs differ and short sales are constrained, that resale option can be valuable even when no investor is intrinsically optimistic about all future dividends.

2 Incremental contribution of the paper

1. Formalizes Keynesian speculation in a rigorous asset-pricing environment. The paper gives a clean formal version of the idea that investors may buy assets not because they personally believe the asset's fundamentals are high, but because they expect to sell later to someone with more optimistic beliefs.

2. Distinguishes dividend value from ownership value. In complete markets with homogeneous or fully arbitrated beliefs, a stock is worth the discounted value of dividends. In this model, the stock is worth the discounted value of dividends *plus* a dynamic resale option.

3. Introduces a consistency criterion for prices with heterogeneous expectations. The paper adapts Radner-style equilibrium with price expectations to a partial-equilibrium stock market. A price scheme is consistent if, given expected future prices, no investor class can obtain excessive expected return through admissible speculative trading.

4. Connects speculative premia, martingales, and bubbles. The minimal consistent price scheme is obtained recursively. Nonminimal consistent price schemes can be interpreted as adding a Ponzi-like speculative bubble to the minimal consistent price.

5. Shows why active trading can be rational. In this model, a buy-and-hold strategy is not generally optimal. Even rational investors may need to actively manage portfolios, buying only in certain states and selling according to particular stopping rules.

3 Page-by-page coverage map

3.1 Table: Coverage of the full paper

Coverage of the full paper	
Pages	Content covered in these notes
1	JSTOR cover page and bibliographic information.
2–4	Introduction: definition of speculative behavior, connection to complete markets, incomplete markets, Keynes, and the paper’s assumptions.
4–5	Formal model: discrete trading dates, information process, dividends, investor classes, subjective expectations, risk neutrality, no short sales.
5–7	Numerical example: two investor classes, two-state dividend process, Markov beliefs, buy-and-hold valuations, and speculative resale logic.
7–11	Consistent price schemes: legitimate selling strategies, consistency condition, one-period Bellman equation, minimal consistent price scheme, nonminimal bubble schemes, and homogeneous-investor benchmark.
11–12	Return to example: explicit computation of minimal consistent prices.
12–13	Relaxing assumptions: class/state-dependent discount factors and a risk-averse interpretation with riskless bonds.
13–15	Concluding remarks: relation to fundamental analysis, technical analysis, efficient markets, and active portfolio management; note and references.

4 Data and measurement: theoretical objects, not empirical data

This is a theoretical paper. It has no empirical sample, no regression design, and no numerical calibration beyond the illustrative two-state example. The key “data” objects are primitives of the model.

4.1 Time, information, and dividends

Trading occurs at discrete dates $t = 0, 1, 2, \dots$. The stock pays a nonnegative dividend d_t at each date $t \geq 1$. New economic information arrives between dates; the total information available by date t is denoted

$$X_t = (\xi_1, \xi_2, \dots, \xi_t),$$

and x_t is a realized history. The dividend d_t is measurable with respect to X_t , so it is written as $d_t(x_t)$.

Interpretation. The state variable is not a single shock, but the accumulated public history. Future prices can depend on the full history because future investors condition their beliefs on realized information.

4.2 Investor classes and subjective expectations

There is a finite set A of investor classes. Investors within the same class share the same subjective probability distribution over the process $\{X_t\}$, while different classes may disagree. The expectation operator for class $a \in A$ is $E^a[\cdot]$.

All classes are assumed risk-neutral and discount at a common rate γ per period. A random payment stream $\{y_{t+s}(X_{t+s})\}_{s \geq 0}$ has value to class a at history x_t equal to

$$E^a \left[\sum_{s=0}^{\infty} \gamma^s y_{t+s}(X_{t+s}) \mid X_t = x_t \right].$$

Economic message. The entire model is driven by disagreement about conditional probabilities, not by different risk aversion, different information sets, or mechanical differences in wealth.

4.3 Short-sale constraint

The paper assumes the stock cannot be sold short. This assumption is crucial. If pessimistic investors could short freely, they would counteract optimistic investors' willingness to pay. With no short sales, pessimists can step aside but cannot push prices down enough to remove speculative premia.

5 Model and empirical specification

5.1 Complete-market benchmark

In a complete set of contingency-claims markets, with unrestricted buying and selling, the stock price equals the arbitrage value of future dividends:

$$P_0 = \sum_t \sum_{x_t} \beta_t(x_t) d_t(x_t),$$

where $\beta_t(x_t)$ is the current price of one dollar paid in contingency x_t at date t .

Benchmark intuition. In that world, reopening markets does not matter. Investors can already choose the entire future-contingent allocation at date zero, so resale optionality adds nothing.

5.2 Speculation definition

The paper defines speculative behavior as follows:

Investors exhibit speculative behavior if the right to resell a stock makes them willing to pay more for it than they would pay if obliged to hold it forever.

This definition is narrow but powerful. It isolates the value of *future resale* from the value of future dividends.

5.3 Equilibrium ownership rule

With risk-neutral investor classes, infinite within-class wealth, and no short sales, all stock is held in each state by the class that gives it the highest value. The price is the maximum value across classes.

Interpretation. The equilibrium price is not the average belief. It is closer to a dynamic sequence of *optimistic valuations*, because pessimistic beliefs are muted by the short-sale constraint.

6 The numerical example

6.1 Set-up

There are two investor classes, $a = 1, 2$, and two dividend states $d \in \{0, 1\}$. Each class believes the dividend follows a stationary Markov chain, but the transition matrices differ:

$$Q^a = \begin{bmatrix} q^a(0, 0) & q^a(0, 1) \\ q^a(1, 0) & q^a(1, 1) \end{bmatrix}.$$

The numerical example uses

$$Q^1 = \begin{bmatrix} 1/2 & 1/2 \\ 2/3 & 1/3 \end{bmatrix}, \quad Q^2 = \begin{bmatrix} 2/3 & 1/3 \\ 1/4 & 3/4 \end{bmatrix}, \quad \gamma = 0.75.$$

6.2 Buy-and-hold values

If investors were forced to buy and hold forever, the expected present values are:

$$\begin{aligned} p^1(0) &= \frac{4}{3} = 1.33, & p^1(1) &= \frac{11}{9} = 1.22, \\ p^2(0) &= \frac{16}{11} = 1.45, & p^2(1) &= \frac{21}{11} = 1.91. \end{aligned}$$

Class 2 values the dividend stream more than class 1 in both states, so a static intuition would say class 2 should always hold the stock.

6.3 Why the static intuition fails

Suppose the state is 0 and the price in state 1 is expected to be 1.91. A class 1 investor may buy in state 0 and sell when state 1 arrives. From class 1's perspective, the expected present value of that trading strategy is

$$\left[\frac{1}{2}(0.75) + \left(\frac{1}{2}\right)^2 (0.75)^2 + \dots \right] (1 + 1.91) = 1.75.$$

This exceeds class 2's buy-and-hold value in state 0 of 1.45.

Economic message. Class 1 does not buy in state 0 because it is more optimistic about long-run dividends. It buys because it expects state 1 to trigger class 2's optimism, allowing resale at a high price.

6.4 Infinite feedback

Once the state-0 price rises to reflect class 1's resale strategy, class 2 may pay more in state 1 because it expects to resell in state 0 later. Thus the two states' prices recursively feed into each other. The equilibrium price is the fixed point of this speculative recursion.

7 Consistent price schemes

7.1 Price schemes and legitimate selling strategies

A price scheme is a sequence of nonnegative functions

$$\{p_0, p_1, p_2, \dots\}, \quad p_t : X_t \rightarrow \mathbb{R}_+.$$

It tells investors what the stock price will be at every possible future information history.

A t -legitimate selling strategy is a stopping time T satisfying $t+1 \leq T \leq \infty$. It means the investor decides when to sell based only on information available at the selling date.

7.2 Consistency condition

A price scheme is **consistent** if, for every date t and history x_t ,

$$p_t(x_t) = \max_{a \in A} \sup_T E^a \left[\sum_{k=t+1}^T \gamma^{k-t} d_k(X_k) + \gamma^{T-t} p_T(X_T) \mid X_t = x_t \right],$$

where the supremum is over all legitimate selling strategies.

Read. The current price must equal the maximum value any investor class can obtain by holding the stock, collecting dividends, and optimally choosing when to sell.

Economic message. The price is the value of an optimal stopping problem. Ownership has option value because the holder can choose the future selling date.

7.3 One-period characterization

Proposition 1 shows that consistency is equivalent to the one-period equation

$$p_t(x_t) = \max_{a \in A} E^a [\gamma d_{t+1}(X_{t+1}) + \gamma p_{t+1}(X_{t+1}) \mid X_t = x_t].$$

Interpretation. If no investor can earn a positive expected excess return by buying today and selling tomorrow, then no admissible multi-period stopping strategy can do better. This uses a discrete optional-stopping logic.

8 Minimal consistent price scheme and speculative bubbles

8.1 Recursive construction

Set $p_t^0 = 0$. For $n = 1, 2, \dots$, define recursively

$$p_t^n(x_t) = \max_{a \in A} E^a [\gamma d_{t+1}(X_{t+1}) + \gamma p_{t+1}^{n-1}(X_{t+1}) \mid X_t = x_t].$$

The sequence $p_t^n(x_t)$ is nondecreasing in n , and its limit is denoted $p_t^*(x_t)$.

8.2 Minimality result

Proposition 2 states that $\{p_t^*\}$ is a consistent price scheme and is minimal: if $\{p_t\}$ is any other consistent scheme, then

$$p_t(x_t) \geq p_t^*(x_t)$$

for all dates and histories.

Economic message. The minimal scheme is the smallest set of prices that prevents every investor class from finding a profitable speculative trading strategy.

8.3 Nonminimal schemes as bubbles

If $\{p_t^*\}$ is not identically infinite, then nonminimal consistent schemes can exist. For example,

$$p_t = p_t^* + c\gamma^{-t}, \quad c > 0,$$

is also consistent. The difference between such a price scheme and the minimal one is a Ponzi-like speculative bubble.

Important distinction. The paper has two layers of price inflation:

- the **speculative premium** in the minimal consistent price, caused by heterogeneous beliefs and resale options;
- the **bubble component** in nonminimal consistent prices, which is an additional self-sustaining Ponzi-like term.

9 Homogeneous-investor benchmark

If there is only one investor class, Proposition 3 shows that the minimal consistent price is simply the expected present value of future dividends:

$$p_t^*(x_t) = E^a \left[\sum_{k=1}^{\infty} \gamma^k d_{t+k}(X_{t+k}) \mid X_t = x_t \right].$$

If another consistent scheme exists, it must be expressible as

$$p_t(x_t) = p_t^*(x_t) + \gamma^{-t} Z_t(x_t),$$

where $\{Z_t\}$ is a nonnegative martingale under that class's beliefs:

$$E^a[Z_{t+1}(X_{t+1}) \mid X_t = x_t] = Z_t(x_t).$$

Economic message. With homogeneous beliefs, speculation disappears. Only dividend fundamentals and possible martingale bubbles remain.

10 Return to the example: minimal consistent prices

The two-state example produces time-independent minimal consistent prices:

$$p^*(0) = \frac{24}{13} \approx 1.85, \quad p^*(1) = \frac{27}{13} \approx 2.08.$$

At these prices, the stock is held by class 1 in state 0 and by class 2 in state 1.

Comparison with buy-and-hold values.

- In state 0, the highest buy-and-hold value is class 2's 1.45, but the consistent price is 1.85.
- In state 1, the highest buy-and-hold value is class 2's 1.91, but the consistent price is about 2.08.

Takeaway. The difference between these numbers is the speculative premium. It is not an irrational bubble; it is the minimal price consistent with rational speculative trading under disagreement and no short sales.

11 Relaxing the assumptions

The paper argues that the analysis can be extended in several directions.

11.1 State- and class-dependent discounting

The common discount factor γ can be replaced by class-specific, state-dependent one-period discount factors $\gamma_t^a(x_t)$. The one-period consistency equation becomes

$$p_t(x_t) = \max_{a \in A} E^a [\gamma_t^a(X_{t+1}) (d_{t+1}(X_{t+1}) + p_{t+1}(X_{t+1})) \mid X_t = x_t].$$

Interpretation. Speculation does not require a constant discount rate. The core logic survives when discounting varies across classes and states.

11.2 Risk aversion and large classes

The paper also argues that risk neutrality and infinite wealth can be interpreted as approximations. If each class has many risk-averse investors and riskless one-period bonds allow intertemporal smoothing, then no individual investor needs to hold enough stock for risk aversion to be first order. In that case, the risk-neutral valuation is a good approximation.

11.3 The nonnegotiable assumption: short-sale constraints

The paper emphasizes that no short sales remain crucial. With perfect short selling, the supply available to optimists would not be fixed; pessimists could expand effective supply through shorts. Equilibrium would require sufficiently different long and short positions so that marginal beliefs align.

12 Figures

12.1 Panels 1–2: Formulation of investor classes, risk-neutral valuation, no short sales, and the two-state numerical example

tors have no control over operations of the firm whose stock is being traded. Thus, they view the dividend stream $\{d_t\}$, and more generally the economic information process $\{\chi_t\}$, as an exogenous source of uncertainty.

We denote by A the finite set of investor classes. Each investor has a subjective probability distribution for the random process $\{\chi_t\}$, and we assume that two members of the same class assess the same probability distribution, although members of different classes may assess different distributions. We denote by $E^a[\cdot]$ an expected value with respect to the probability distribution shared by members of class $a \in A$.

All investors are assumed to be risk-neutral, discounting future income at rate γ per period. At time t , in contingency x_t , an investor of class a is thus indifferent between a random stream of payments

$$\{y_{t+s}(\chi_{t+s}); s = 0, 1, \dots\}$$

and a certain payment in the amount

$$E^a \left[\sum_{s=0}^{\infty} \gamma^s y_{t+s}(\chi_{t+s}) | \chi_t = x_t \right].$$

Here $E^a[\cdot | \chi_t = x_t]$ means conditional expectation, given that $\chi_t = x_t$. Each class of investors is assumed large enough to prevent collusion and wealthy enough to buy up all of the stock if it so desires.

Our key assumption is that the stock in question cannot be sold short. This is necessary to prevent our infinitely wealthy investor

Formulation: investor classes, risk-neutral valuation, and no short sales.

by $q^a(d, d')$ the probability assessed by class a for a transition from state d to state d' . We define the transition matrices,

$$Q^a = \begin{bmatrix} q^a(0, 0) & q^a(0, 1) \\ q^a(1, 0) & q^a(1, 1) \end{bmatrix} \quad \text{for } a = 1, 2,$$

and assume the specific numerical data,

$$Q^1 = \begin{bmatrix} 1/2 & 1/2 \\ 2/3 & 1/3 \end{bmatrix}, \quad Q^2 = \begin{bmatrix} 2/3 & 1/3 \\ 1/4 & 3/4 \end{bmatrix},$$

and

$$\gamma = 0.75.$$

Note that each class is positive that it knows the actual transition matrix, so that no amount of transition frequency data will alter their assessments. This is implicit in the statement that only the current state of the market is judged relevant.

Beginning the search for equilibrium prices, we compute the value to the investors of buying the stock and holding it forever. Given their assumed preference structure, this is the expected present worth of future dividends. Denote by $p^a(d)$ the expected present worth of future dividends to an investor of class a when the state of the market is d . Simple calculations give

$$\begin{aligned} p^1(0) &= 4/3 = 1.33, & p^1(1) &= 11/9 = 1.22, \\ p^2(0) &= 16/11 = 1.45, & p^2(1) &= 21/11 = 1.91. \end{aligned}$$

It is interesting to note that members of class 2 assess a higher expected present worth of future dividends than do members of class 1, regardless of the current state. One might therefore conjecture that members of class 2 will always hold the stock and that the price in each state d will be $p^2(d)$.

But if members of class 1 anticipate these prices, the market Example: transition matrices, discount factor, and buy-and-hold values.

Read. The first panel defines the valuation environment. The second panel shows why a purely fundamental buy-and-hold approach is insufficient in the two-state example.

Economic message. Heterogeneous beliefs and the no-short-sale constraint make the stock's owner identity state-dependent: the holder in one state may buy partly to sell to a different holder in another state.

12.2 Panels 3–4: Consistent price schemes and the one-period characterization in Proposition 1

event $\{U = t + k\}$ corresponding to the set of all circumstances that would cause the investor to sell the stock at time $t + k$. The requirement that T be optional with respect to $\{\chi_t\}$ simply insures that the choice of whether or not to sell at time $t + k$ will be based solely on the information χ_{t+k} that is then available. Finally, we shall say that a price scheme $\{p_0, p_1, \dots\}$ is *consistent* if

(1)

$$p_t(x_t) = \max_{a \in A} \sup_T E^a \left[\sum_{k=t+1}^T \gamma^{k-t} d_k(\chi_k) + \gamma^{T-t} p_T(\chi_T) \mid \chi_t = x_t \right]$$

for all $t = 0, 1, \dots$ and all $x_t \in X_t$, the supremum being over all t -legitimate selling strategies. Our argument that (1) is a natural condition for equilibrium in the market goes as follows. Suppose that a price scheme $\{p_t(\cdot)\}$ is to be followed. Then for each class $a \in A$, the expression,

Consistency condition: price equals the best value over classes and selling rules.

As we have seen in our example, the achievement of an equilibrium entails a delicate interplay among the subjective assessments of the various classes, so it is not immediately obvious that consistent price schemes always exist. To show that they do exist, we first simplify the definition of consistency. Because dividends and price schemes are nonnegative by assumption, a standard result in discrete potential theory allows us to drop the supremum over all selling strategies and consider only strategies that buy, hold for one period, and then sell.

PROPOSITION 1. A price scheme $\{p_t\}$ is consistent if and only if, for all t and x_t ,

$$(2) \quad p_t(x_t) = \max_{a \in A} E^a [\gamma d_{t+1}(\chi_{t+1}) + \gamma p_{t+1}(\chi_{t+1}) \mid \chi_t = x_t].$$

The proof is as follows. Suppose that (2) holds. Then for all $a \in A$, $p_t(x_t) \geq E^a [\gamma d_{t+1}(\chi_{t+1}) + \gamma p_{t+1}(\chi_{t+1}) \mid \chi_t = x_t]$, so speculation is viewed by all investors as no better than a “fair” game. Doob’s optional stopping theorem (cf. Chung, 1974) implies that no nonanticipatory strategy for speculation can do better than “break even.” (Because d_t and p_t are nonnegative, unbounded, and even infinite, stopping times are permitted.) That is,

Proposition 1: the optimal-stopping condition reduces to a one-period equation.

Read. The first panel states the full consistency definition with a supremum over legitimate selling strategies. The second panel shows the Bellman-type simplification.

Economic message. Consistent prices are not simply expected dividends; they are fixed points of the dynamic resale option.

12.3 Panels 5–6: Recursive minimal consistent price scheme and homogeneous belief benchmark

do so at some point using a simple strategy that buys, holds for one period, and then sells. (This is not to say that strategies that hold the stock for many periods will always achieve an expected net present value of zero when prices are consistent. See the discussion in Section VII.)

To identify explicitly a consistent price scheme, set $p_t^0 = 0$ and for $n = 1, 2, \dots$, recursively define

$$(3) \quad p_t^n(x_t) = \max_{a \in A} E^a [\gamma d_{t+1}(\chi_{t+1}) + \gamma p_{t+1}^{n-1}(\chi_{t+1}) \mid \chi_t = x_t].$$

Observe that $p_t^n(x_t)$ is nondecreasing in n and hence approaches some (possibly infinite) limit $p_t^*(x_t)$ as $n \rightarrow \infty$.

PROPOSITION 2. The price scheme $\{p_t^*\}$ is consistent. If $\{p_t\}$ is any other consistent scheme, then $p_t(x_t) \geq p_t^*(x_t)$ for all t and x_t . That is, $\{p_t^*\}$ is the minimal consistent price scheme.

A sketch of the proof follows. To show that $\{p_t^*\}$ satisfies (2) and hence is consistent, let $n \rightarrow \infty$ in equation (3). On the right-hand side,

Recursive construction of the minimal consistent price scheme.

PROPOSITION 3. If there is a single investor class a , then the minimal consistent price scheme $\{p_t^*\}$ is the expected present worth of future dividends (for that class). Moreover, if $\{p_t^*\}$ is finite, another price scheme $\{p_t\}$ is consistent if and only if

$$p_t(x_t) = p_t^*(x_t) + \gamma^{-t} Z_t(x_t),$$

where $\{Z_t\}$ is a nonnegative martingale with respect to $\{\chi_t\}$ and the investors’ probability assessments, meaning that

$$E^a [Z_{t+1}(\chi_{t+1}) \mid \chi_t = x_t] = Z_t(x_t) \quad \text{for all } t \text{ and } x_t.$$

Proof. Inductively,

$$p_t^n(x_t) = E^a \left[\sum_{k=t}^n \gamma^k d_{t+k} \mid \chi_t = x_t \right],$$

Homogeneous-belief benchmark and martingale representation.

Read. The recursive construction begins from zero continuation value and iterates forward. The homogeneous-investor benchmark collapses the model back to fundamental value plus possible martingale bubbles.

Economic message. Heterogeneous beliefs are the source of the minimal speculative premium; homogeneous beliefs remove it.

12.4 Panels 7–8: Minimal consistent prices in the example and class/state-dependent discounting

so by monotone convergence

$$p_t^*(x_t) = E^a \left[\sum_{k=1}^{\infty} \gamma^k d_{t+k} | \chi_t = x_t \right].$$

The characterization of nonminimal price schemes follows by subtracting equation (2) for $\{p_t^*\}$ from equation (2) for any other consistent $\{p_t\}$.

V. BACK TO THE EXAMPLE

The stationary nature of our example guarantees that the minimal consistent price scheme $\{p_t^*\}$ will be time-independent. That is, it consists simply of prices $p^*(0)$ and $p^*(1)$ for states zero and one, respectively. Thus (2) can be rewritten as

$$\begin{aligned} p^*(0) &= \max\{(\frac{3}{4})(\frac{1}{2})p^*(0) + (\frac{3}{4})(\frac{1}{2})(1 + p^*(1)), \\ &\quad (\frac{3}{4})(\frac{2}{3})p^*(0) + (\frac{3}{4})(\frac{1}{3})(1 + p^*(1))\} \\ p^*(1) &= \max\{(\frac{3}{4})(\frac{2}{3})p^*(0) + (\frac{3}{4})(\frac{1}{3})(1 + p^*(1)), \\ &\quad (\frac{3}{4})(\frac{1}{4})p^*(0) + (\frac{3}{4})(\frac{3}{4})(1 + p^*(1))\}, \end{aligned}$$

for which the unique solution is $p^*(0) = 24/13 = 1.85$ and $p^*(1) = 27/13 = 2.04$. At these prices the stock is held by class 1 investors in state zero and by class 2 investors in state one. In Section III we described a (monotonically increasing) price iteration procedure of the

Back to the example: minimal consistent prices in states 0 and 1.

Read. The example produces explicit prices $p^*(0) = 24/13$ and $p^*(1) = 27/13$. The extension shows that the same logic works with more general discounting.

Economic message. The numerical solution demonstrates how resale value raises prices above every class's own buy-and-hold dividend valuation.

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with $p_t^0 \equiv 0$. This iterative procedure is different from that used to define p_t^* in Section IV, but they can be shown in general to yield the same limit as $n \rightarrow \infty$.

VI. RELAXING THE ASSUMPTIONS

For each $a \in A$, let $\{\gamma_t^a\}$ be a sequence of nonnegative functions such that γ_t^a has domain X_t . Let $\{y_t, y_{t+1}, \dots\}$ be a random stream of payments as in Section II. Our entire treatment is easily generalized to the case where members of class a , observing contingency x_t at time t , value the stream $\{y_{t+s}\}$ at

$$E^a \left[\sum_{s=0}^{\infty} \left(\prod_{k=1}^s \gamma_{t+k}^a(\chi_{t+k}) \right) y_{t+s}(\chi_{t+s}) | \chi_t = x_t \right].$$

That is, we can allow for one period discount rates that are particular to the investor's class and also depend on the circumstances in which the income being discounted is received. Our analysis goes through without a hitch. In particular, (2) becomes

$$p_t(x_t) = \max_{a \in A} E^a [\gamma_{t+1}^a(\chi_{t+1})(d_{t+1}(\chi_{t+1}) + p_{t+1}(\chi_{t+1})) | \chi_t = x_t].$$

Relaxing assumptions: class/state-dependent discount factors.

12.5 Panel 9: Concluding discussion on active portfolio management and the efficient-market hypothesis

Proponents of the efficient market hypothesis conclude that the rational portfolio strategy (in view of transaction costs and risk aversion) is to buy a well-diversified portfolio and hold it. Quite a different view of “rational” portfolio management emerges from our model. Consider again the example presented in Section III. Investors of class 1 cannot achieve an expected net present value of zero from stock bought in state one, and they can achieve this from stock bought in state zero if and only if they sell at or before the first time a one dollar dividend is declared. In the general model, investors can achieve an expected net present value of zero only from stock bought in certain circumstances and only if they follow certain selling strategies. (The strategy of selling after one period, which leads to much churning of the portfolio, always works.) The strategy of buying in favorable circumstances and holding for many periods typically yields an expected loss. In brief, all investors must actively manage their portfolios in order to expect a proper return.

If our model were broadened to include risk aversion and transaction costs, there would be forces at work that yield a notion of “rational” portfolio management more in accord with the efficient market hypothesis. But the conflict between our model and the hypothesis lies at a deeper, philosophical level. As we understand things, a subjectivist statement of the hypothesis implicitly requires the supposition that rational economic agents having the same information will arrive at the same subjective probability assessments. The hypothesis then says that, although investors do not originally have the same information, prices reflect their information in such a way as

Concluding discussion: active portfolio management and the efficient-market hypothesis.

Read. The final panel explains why the model conflicts with a strong subjectivist version of the efficient-market hypothesis: even with the same substantive information, investors may retain different subjective assessments.

Economic message. Rational trading need not be passive buy-and-hold. If resale values depend on future disagreement, a rational investor must think about future market beliefs and selling strategies.

13 Interpretation of the theory

13.1 Why prices can exceed all hold-forever values

The stock’s value to an investor class is not only

dividend value,

but rather

dividend value while held + resale value at the chosen stopping time.

If future resale value is high in certain states, an investor may bid up the current price even when she assigns low value to the long-run dividend stream.

13.2 Why short-sale constraints matter

The no-short-sale constraint turns disagreement into asymmetric pricing power. Optimists can buy and hold all supply; pessimists can only decline to buy. Therefore, the market price reflects the optimistic valuation of the marginal potential holder, not the average belief across all agents.

13.3 Why future market reopening matters

If the market never reopened, there would be no resale option. Investors would value the stock as a dividend stream. Once the market reopens, the price today reflects expected future opportunities to sell to other investor classes.

13.4 Why the model is not merely a bubble model

The minimal consistent price already contains a speculative premium. That premium is generated by heterogeneous beliefs and resale value. Additional nonminimal price schemes are bubble-like, but the speculative premium itself is present even in the minimal equilibrium.

14 How to read the paper as an asset-pricing contribution

Relation to standard fundamental value. The paper does not deny that dividends matter. Instead, it shows that when markets are incomplete and investors disagree, the relevant market aggregation of beliefs can value ownership of the stock more than ownership of the dividend stream alone.

Relation to market efficiency. The paper questions a version of efficiency in which common information leads rational agents to the same beliefs. Here agents have the same substantive information but use different subjective probability assessments; prices therefore aggregate disagreement rather than reveal a single common posterior.

Relation to modern disagreement models. The paper is an early foundation for models in which heterogeneous beliefs and short-sale constraints generate overpricing, speculative trading, and bubbles. Later work on disagreement, resale options, and limits to arbitrage builds directly on this logic.

15 Limitations and assumptions

- **Partial equilibrium.** Dividends, information arrival, discounting, and other prices are taken as exogenous.
- **Perfect contingent foresight.** Investors are assumed to have a common, correct conception of the entire future price scheme in every contingency.
- **Risk neutrality and wealth assumptions.** The baseline model assumes risk-neutral investor classes with enough wealth to buy the whole stock supply.
- **No short sales.** This is essential. Relaxing it substantially changes the mechanism.

- **No endogenous information production.** Investors disagree in beliefs, but the model does not explain why the disagreement persists or how information acquisition evolves.
- **No transaction costs.** The active-management conclusion would be softened by transaction costs and risk aversion.

16 Short summary

Problem. In complete markets with homogeneous beliefs, stock prices equal discounted dividends. But real markets are incomplete, investors disagree, and short-sale constraints restrict pessimists.

Mechanism. Because the stock can be resold, an investor may buy it in order to sell later to another investor who will become more optimistic in a future state.

Mathematical object. A consistent price scheme is a fixed point of an optimal-stopping valuation problem across all investor classes.

Result. The minimal consistent price can exceed every investor class's expected present value of dividends. The excess is a speculative premium. Nonminimal consistent prices add bubble-like martingale/Ponzi components.

Takeaway. Speculation can be rational, not because investors are irrationally optimistic, but because future disagreement and resale opportunities are valuable.

17 Conclusion

17.1 Core conclusion

Harrison and Kreps (1978) shows that, under heterogeneous expectations and short-sale constraints, a stock can trade at a price above the dividend valuation of every investor class. This occurs because the asset includes a resale option. Investors price not only cash flows but also future opportunities to sell the asset to someone else.

17.2 Incremental contribution revisited

The paper's deep contribution is to make speculative resale value a rigorous equilibrium object. Its minimal consistent price is neither a naive fundamental value nor necessarily an irrational bubble. It is the smallest price process that prevents any investor class from earning excessive expected returns through legitimate speculative trading.

17.3 Economic intuition

The model formalizes a simple but powerful idea: disagreement plus resale opportunities can create value. If future buyers will interpret future news differently from current buyers, the current stock price must capitalize the option to sell to those future buyers.

17.4 What to remember

- The price of the stock can exceed the value of the dividend stream.
- Heterogeneous expectations alone are not enough; the resale option and short-sale limits are central.
- Consistency is a dynamic fixed point over investor beliefs and future prices.
- Homogeneous beliefs collapse the model back to fundamental value plus possible martingale bubbles.
- The model provides a theoretical foundation for speculative premia and active trading under disagreement.